Yoel Kim

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Research Interests

- Software Engineering (SE) → software verification; specification mining; embedded software.
- Formal Methods (FM) → abstraction; model checking; automata learning.
- Programming Languages (PL) → program analysis; program synthesis.

Education

Ph.D. in Computer Science and Engineering, Kyungpook National University — *Mar 2023 – Feb 2027*
- Advisor: Yunja Choi
- Dissertation: Data-Driven Abstractions for Embedded Software

M.S. in Computer Science and Engineering, Kyungpook National University — *Mar 2021 – Feb 2023*
- Advisor: Yunja Choi
- Thesis: An Automated Stub Generation Approach Using Program Synthesis for Software Verification

B.S. in Computer Science and Engineering, Kyungpook National University — *Mar 2017 – Feb 2021*
- Track: Global Software Convergence
- GPA: 3.95/4.3, Rank: 1/13

International Conferences

1. AutoCL: An End-to-End Tool for Active Automata Learning from C Programs
   Yoel Kim, Yong-Hwan Jeong, and Yunja Choi
   TACAS'27: *International Conference on Tools and Algorithms for the Construction and Analysis of Systems*
   Major FM Conference | **BKCS IF: 2** | **Regular Tool Papers Track** | **Under Review**

2. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper
   Yoel Kim and Yunja Choi
   FSE'26: *ACM International Conference on the Foundations of Software Engineering*
   Top-tier SE Conference | **BKCS IF: 4** | **Acceptance Rate: 22.9% (211/920)**

3. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software
   Yoel Kim and Yunja Choi
   FSE'24: *ACM International Conference on the Foundations of Software Engineering*
   Top-tier SE Conference | **BKCS IF: 4** | **Direct Acceptance Rate: 11.6% (56/483)**

International Journals

1. Learning Functional Specifications from Stateful Control Programs through Active Model Learning
   Yunja Choi and Yoel Kim
   IEEE Access, 2026
   SCIE | **IF: 4.2** | **Q2**

Domestic Conferences

1. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation
   Seungbin Choi, Yoel Kim, and Yunja Choi
   KCSE'26: *Korea Conference on Software Engineering* ** Short Paper Award*

2. An Approach of Incremental Constraint Extraction Based on I/O Examples for Automatic Stub Generation
   Yoel Kim and Yunja Choi
   KCSE'23: *Korea Conference on Software Engineering* ** Best Short Paper Award*

3. A Case Study to Improve the Efficiency of Model Checking in Embedded Software Using Program Synthesis
   Yoel Kim and Yunja Choi
   KSC'21: *Korea Software Congress*

4. A Case Study on the Performance of Program Synthesis in Embedded Software Domain
   Yoel Kim and Yunja Choi
   KCSE'21: *Korea Conference on Software Engineering*

Domestic Journals

1. An Analysis of the Feasibility and Limitations of LLM-based State Machine Generation for Reactive Systems
   Seungbin Choi, Yoel Kim, and Yunja Choi
   KIPS Transactions on Software and Data Engineering, 2026

Selected Talks

1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper. Research Paper Presentation. FSE'26. *Concordia University, Montreal, Canada. Jul 7, 2026.*

2. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCC'25. *Jeju, Korea. Jul 4, 2025.*

3. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference Session. KCSE'25. *Pyeongchang, Korea. Jan 22, 2025.*

4. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Research Paper Presentation. FSE'24. *Porto de Galinhas, Ipojuca, Pernambuco, Brazil. Jul 17, 2024.*

Honors and Awards

- Short Paper Award, Korea Conference on Software Engineering (KCSE), 2026
- Best Short Paper Award, Korea Conference on Software Engineering (KCSE), 2023
- National Scholarship for Science and Engineering, 2019–2020

Teaching Experience

Teaching Assistant, Kyungpook National University *Mar 2021 – Jun 2024*
- ITEC0414: Software Testing Theory (Spring 2022, 2023, 2024).
- COMP0224: Software Design (Fall 2021, 2022).
- COMP0216: Data Structure Applications (Spring 2021).

Undergraduate Tutor, Kyungpook National University *Mar 2018 – Dec 2020*
- COMP0322: Database Management Systems (Fall 2020).
- CLTR266: Software Computational Thinking (Spring 2018, 2019).
- CLTR268: Python Programming (Winter 2018).
- COMP0216: Data Structure Applications (Fall 2018).